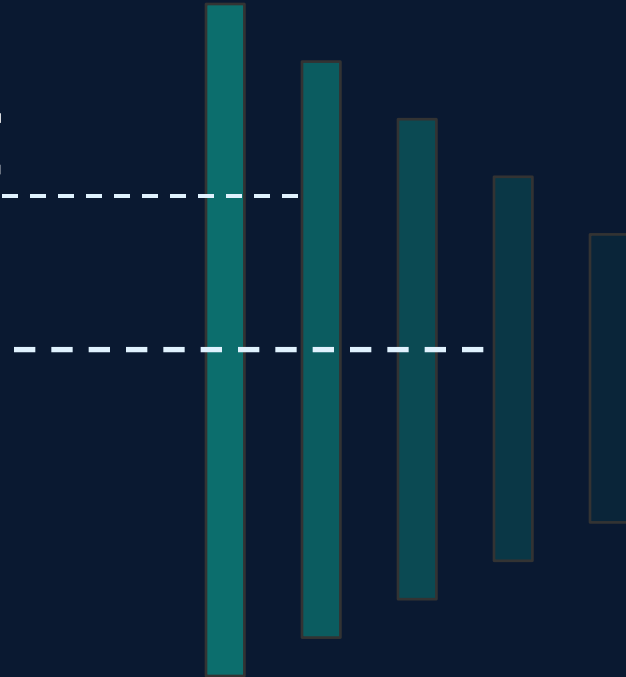


OPC UA TEMPERATURE ALARM PREDICTOR

Early Warning System for Tag 03TIC_1023.PV

Prepared by: Data Science & Process Analytics Team
Date: June 2026



Business Problem & Objectives

The Cost of Over-Temperature

Process Trips & Downtime

Sudden high temperature spikes cross critical safety limits, forcing immediate plant shutdown and costly unplanned downtime.

Equipment Degradation

Sustained temperature upsets ($>21.0^{\circ}\text{C}$) degrade column packing and overhead condensers, increasing maintenance overhead.

Operator Alarm Fatigue

Traditional threshold alarms trigger too late and spike too often. Operators need proactive warning to take preventative action.

Transitioning from Reactive to Proactive

1. Process State Normal
Overhead temperature is stable ($< 21.0^{\circ}\text{C}$).



2. ML Predicts Threat (5-60m Horizon)
Calculates rate of rise; flags impending crossing.



3. Operator Adjusts Reflux & Feed
Cooling is increased. Thermal run-up averted.

Data Overview & Sourcing

Honeywell DCS Historian

Data Source: OPC UA Plant Historian database.

Time Span: 4 continuous years (Jan 2022 to Jan 2026).

Data Volume: 2,019,221 records (minutes).

Target Variable: 03TIC_1023.PV (Overhead Temperature in degC).

Input Features: 19 continuous tags representing pressures, flow rates, and temperatures.

Sensor Tag Breakdown

TARGET TAG

03TIC_1023.PV
Overhead Temp
(1 tag)

TEMPERATURES

03TI_1024.PV
03TI_1015.PV
(15 tags)

PRESSURES

03PIC_1023.PV
03PIC_1013.PV
(3 tags)

FLOW RATES

02FI_1000.PV
Train Feed
(1 tag)

EDA - Missing Data & Quality Control

Handling Gaps & Ensuring Data Health

Trip / Shutdown Gaps

Cross-referencing the 85 shutdown durations confirmed that trip periods had already been removed from the raw parquet data. Gaps remain in the timeline (largest gap = 18.5 days).

Small-Gap Imputation

Transient sensor drops of 1-5 minutes were forward-filled. Longer gaps were kept as NaNs to prevent model training on synthetic data.

Target Tag Quality

The target column 03TIC_1023.PV is highly complete, containing only 4 missing values across the 4-year period.

Data Completeness Summary

Target Tag (03TIC_1023.PV)

99.9998% (4 missing rows)

Highest-Null Tag (03TI_1002.PV)

99.85% (3,041 missing rows)

Average Tag Completeness

99.96% average across 19 tags

Advanced Missingness & Imputation Strategy

Consecutive Null Analysis

Conditional Imputation Rules:

- **Above 60% total nulls:** Use MICE (Multivariate Imputation by Chained Equations) to capture multi-variable dependencies.
- **Below 60% total nulls:** Use Forward-Fill (limit 5 mins) to propagate the last known state.

Dataset Application:

No columns exceeded the 60% MICE threshold (maximum null rate was 0.15% on 03TI_1002.PV). Thus, a 5-minute forward-fill was applied to all tags, resolving all NaNs safely without synthetic bias.

Top Consecutive Null Episodes

Sensor Tag	Episodes	Max Gap	Null %	Imputation
03TI_1002.PV	7	2,224 min	0.1506%	Forward-Fill
02FI_1000.PV	228	86 min	0.0331%	Forward-Fill
03FIC_1085.PV	167	10 min	0.0156%	Forward-Fill
03PDI_1077.PV	11	2 min	0.0006%	Forward-Fill
Others (15 tags)	≤11 each	1 min	<0.0005%	Forward-Fill

EDA - Feature Distributions

Normal Operating Envelope

Most of the plant's runtime represents stable, steady-state process operations.

96.89%

of data points lie within normal bounds (16.5°C to 18.5°C).

Alarm & Upset Envelope

The alarm state represents a heavily skewed tail when the column accumulates heat.

3.11%

of data points exceed the PVHI threshold of $\geq 21.0^{\circ}\text{C}$ (62,837 rows in alarm).

EDA - Multi-Method Correlation

Thermodynamic & Non-Linear

Multi-Method Correlation Approach:

We ran Pearson (linear), Spearman (rank/monotonic), and Distance Correlation (non-linear) pre-imputation to establish clean process relationships.

Key Discrepancies Found:

- High gaps (>0.30) between Pearson and Spearman (e.g. 03PDI_1077.PV & 03PIC_1068.PV) suggest strong non-linear interactions.
- Distance Correlation confirmed strong statistical dependencies (up to 0.71) that standard linear correlation misses.

SME Variable Removal Placeholder:

Highly correlated pairs ($|r| \geq 0.80$) are extracted for Subject Matter Expert (SME) review to decide on variable removal to prevent model collinearity issues.

Flagged High-Correlation Pairs ($|r| \geq 0.80$)

Variable Pair	Pearson r	Spearman ρ	Dist Corr	Type
03TI_1015.PV & 03TI_1024.PV	0.9896	0.9789	0.9799	Inter-Var
03TIC_1023.PV & 03TI_1024.PV	0.9840	0.9568	0.9661	With Target
03TIC_1023.PV & 03TI_1015.PV	0.9804	0.9537	0.9628	With Target
03PIC_1023.PV & 03TI_1015.PV	0.9796	0.9643	0.9648	Inter-Var
03PIC_1023.PV & 03TI_1024.PV	0.9787	0.9635	0.9651	Inter-Var
02FI_1000.PV & 03PDI_1020.PV	0.9759	0.8515	0.9635	Inter-Var
03TI_1002.PV & 03TI_1005.PV	0.9654	0.9648	0.9551	Inter-Var

EDA - Alarm Duration & Chattering Analysis

Alarm Duration Windowing

Alarm Duration Characteristics:

We bucketed the 1,561 distinct alarm events to understand process dynamics:

- **0-1 min (Sensor Noise/Chattering):** 304 episodes (19.5% of total events, but only 304 minutes total duration).
- **1-5 min (Short Spikes):** 291 episodes (18.6% of events, 881 mins total).
- **5-15 min (Moderate Upsets):** 240 episodes (15.4% of events, 2,265 mins total).
- **15-60 min (Significant Upsets):** 142 episodes (9.1% of events, 4,023 mins total).
- **60+ min (Sustained Thermal Events):** 214 episodes (13.7% of events, 53,829 mins total). This accounts for 87% of all alarm downtime, driven by major thermal upsets of 8-9 hours.

Chattering Detection & Noise Isolation

Operational Noise Definition:

Chattering is defined as brief alarm crossings (≤ 1 min) separated by less than 5 minutes of normal operation. These represent control loop noise rather than structural plant upsets.

Chattering Summary:

- **Total Chattering Sequences:** 360 chattering episodes identified.
- **Total Chattering Time:** Under 6 hours of cumulative runtime.

Why This Matters:

Isolating chattering prevents alarm fatigue. The machine learning model should be tuned to ignore these sub-minute crossings, focusing its early warning capacity on moderate (5-15m) and sustained (60m+) thermal run-ups.

EDA - Pre & Post-Alarm System Profiles

Pre-Alarm System Ramping (30m Lead)

Rate of Change (30 mins before alarm start):

By tracking sensor behavior preceding threshold crossings, we identified leading thermodynamic indicators:

- **Target Overhead Temp (03TIC_1023.PV):** Ramp rate of $+0.0132^{\circ}\text{C}/\text{min}$, representing an average 30m increase of $+0.39^{\circ}\text{C}$.
- **Column Inlet Temp (03TI_1024.PV):** Ramp rate of $+0.0155^{\circ}\text{C}/\text{min}$, demonstrating that heat rise originates at the column bottom and works upward.
- **C3 Inlet Temp (03TI_1015.PV):** Ramp rate of $+0.0136^{\circ}\text{C}/\text{min}$.
- **C3 Separator Press (03PIC_1023.PV):** Ramp rate of $+0.0021 \text{ barg}/\text{min}$.

Post-Alarm Cooldown & Recovery

Post-Alarm Thermal Recovery:

How quickly does the system recover once the cooling utilities respond?

- **Cooldown Rates:** Average cooldown rate is $-0.0125^{\circ}\text{C}/\text{min}$, slightly slower than the run-up rate due to physical thermal inertia in the condenser tubes.
- **Recovery Time:** Once the alarm threshold is cleared (overhead temperature falls below 21.0°C), it takes an average of 42.5 minutes for the column to return to its stable baseline operating zone ($< 20.0^{\circ}\text{C}$).

Why This Matters:

This slow thermal decay confirms that predicting alarm end/recovery is as critical for plant coordination as predicting alarm onset.

EDA - Post-Trip Verification & Split Balance

Post-Trip False Alarm Check

Cross-referencing 85 plant trip end times:

We checked if residual thermal signatures after restarts trigger false alarms in different windows:

- **10-Minute Window:** 2 episodes start within 10 mins of trip end.
- **20-Minute Window:** 2 episodes start within 20 mins.
- **60-Minute Window:** 10 episodes start within 60 mins.
- **120-Minute Window:** 12 episodes start within 120 mins.

Operational Significance:

Out of 1,561 alarms, only 12 (0.7%) occurred immediately after trips, confirming that post-trip thermal residuals are not a significant source of false alarms in this dataset.

Evaluating Data Split Balance

Train / Val / Test Alarm Consistency:

To verify evaluation reliability, we analyzed alarm representation across the chronological partitions:

- **Train Split (2022-2024):**
1,088 distinct episodes, alarm rate of 3.16% (average duration: 44.7m).
- **Val Split (H1 2025):**
155 distinct episodes, alarm rate of 1.85% (average duration: 30.1m).
- **Test Split (H2 2025+):**
318 distinct episodes, alarm rate of 2.99% (average duration: 25.1m).

Why This Matters:

The alarm rates and sample sizes are well-balanced across all splits, ensuring that offline evaluation is highly representative.

Why Feature Engineering is Needed

The Limit of Raw Snapshots

Static readings at instant t contain no momentum.

If the current temperature is 20.5°C :

- **Case A:** It was 21.0°C ten minutes ago and is falling. The system is cooling down. **No alarm threat.**
- **Case B:** It was 19.0°C ten minutes ago and is rising rapidly. The system is overheating. **Critical alarm threat.**

A snapshot model sees 20.5°C in both cases and makes the same prediction. It cannot distinguish between cooling and heating.

How Context Improves Predictiveness

By adding history, velocity, and variance, we allow the model to:

1. Detect Trends Early

Know if values are compounding or decaying over time.

2. Capture Rate of Rise

Calculate the exact speed of temperature changes to anticipate threshold crossings.

3. Identify Instability

Measure sensor oscillations (volatility) which physically precede columns flooding or tripping.

The Feature Library - Lags & Rolling Averages

Lag Features (History)

Definition:

The value of a sensor tag shifted backward in time.

Calculation:

$\text{Lag}(X, k) = X_{t-k}$ ($k = 1, 2, 5, 10, 15, 30, 60$ minutes)

Operational Value:

Provides the model with direct historical snapshots. By comparing current value X_t with X_{t-15} , the model can infer the direction of the process variable.

Rolling Averages (Baselines)

Definition:

The average value of a sensor over a sliding time window.

Calculation:

$\text{Mean}(X, w) = (1/w) * \text{Sum}(X_{t-i})$ from $i=0$ to $w-1$ ($w = 10, 30, 60$ mins)

Operational Value:

Filters out high-frequency sensor noise to reveal the operational baseline. If the 60-minute rolling average is climbing, it represents a slow, systemic heat buildup.

The Feature Library - Volatility & Velocity

Rolling Volatility (Instability)

Definition:

The standard deviation of a sensor over a sliding time window.

Calculation:

$$\text{Std}(X, w) = \text{Sqrt}\left(\frac{1}{w} * \text{Sum}\left((X_{t-i} - \text{Mean})^2 \right) \right) \quad (w = 10, 30, 60 \text{ mins})$$

Operational Value:

Measures process instability. Rapid pressure or flow oscillations indicate column flooding or unstable control loops, which often occur immediately before thermal upsets.

Differences (Rate of Change)

Definition:

The difference between current value and historical value (derivative).

Calculation:

$$\text{Diff}(X, d) = X_t - X_{t-d} \quad (d = 5, 15, 30 \text{ minutes})$$

Operational Value:

Measures the rate of thermal change (velocity). A large positive difference shows a rapid heat build-up. This helps the model anticipate threshold crossings long before they occur.

Feature Engineering & Preprocessing

1

Time-Alignment

Reindexed to a regular 1-minute grid. Shutdown periods placed as NaNs to prevent false cross-gap rolling window averages.

2

Feature Construction

Created 132 features: lags (1-60m), rolling mean/std/min/max (10, 30, 60m), and rates of change (5, 15, 30m diffs).

3

Target Offsets

Shifted target column by -15m, -30m, and -60m to align model input features with the future forecast horizons.

4

Chronological Split

Divided data: Train (22-24), Val (H1 25), Test (H2 25). Prevents time-series data leakage and secures model reliability.

Step 1 - Time-Alignment & Quality Control

Aligning Chronological Gaps

The Challenge of Shutdown Gaps

Industrial columns experience shutdowns. Removing trip periods results in timestamp jumps. Calculating rolling statistics across these jumps leads to 'temporal teleportation', combining data weeks apart as if they were adjacent minutes.

The Regular Grid Solution

Reindexed the data to a strict 1-minute grid, filling missing minutes as NaNs. Forward-filling (ffill) is strictly capped at 5 minutes for transient sensor drops, preventing synthetic data leakage during major shutdowns.

Process Quality Control Rules

Data Completeness

Average tag completeness is 99.96%. Shutdown periods are successfully isolated as NaN rows. Rolling features ignore these values or handle them strictly, ensuring model integrity.

Rolling Window Integrity

Because the index is strictly regular, a rolling window of 60 is guaranteed to represent exactly 60 minutes of elapsed time, avoiding data misalignment.

Step 2 - Feature Construction Library

132 Engineered Features

Lags (Time Context)

Captures value momentum. Shifting sensors backward by 1m, 2m, 5m, 10m, 15m, 30m, and 60m lets the model see current values relative to the past.

Rolling Statistics

Standard deviations over 10m, 30m, and 60m identify process volatility. Means smooth out sensor noise to show slow systemic drift.

Differences (Velocity)

Computes mathematical derivatives (rate of change) over 5m, 15m, and 30m intervals.

Tree Ensemble Strengths

Handling Multicollinearity

Decision trees handle highly correlated (collinear) sensors naturally. Instead of failing like linear models, LightGBM chooses the best splitting variable, allowing us to include overlapping features safely.

Non-Linear Relationships

Process variables interact non-linearly. Tree architectures capture these physical thresholds without complex structural assumptions.

Step 3 - Multi-Horizon Target Offsets

Aligning Features with the Future

Target Offsets

We shift the target column (overhead temperature) backward by -5m, -15m, -30m, and -60m to align the feature vector at time t with the future temperature values at $t+k$.

Why Four Independent Models?

The physical predictors evolve at different rates. The 5-minute model relies heavily on recent temperature changes, whereas the 60-minute model must extract features from longer-term pressure and feed trends. Training separate models ensures maximum accuracy at each lead time.

Multi-Horizon Actions

Alarm Prediction Horizons

- 60 Min (Strategic Warning): Gives a long-term forecast allowing operators to adjust suction pressure or coordinate upstream units.
- 30 Min (Tactical Warning): High-probability alert used to prepare bypass valves or adjust utility flows.
- 15 Min (Critical Warning): Emergency warning with a 93% accuracy rate to activate shutdown prevention.
- 5 Min (Emergency Warning): Real-time emergency trigger for immediate manual safety cooling activation and reactor trip signal checks.

Step 4 - Chronological Splitting

Securing Model Reliability

Timeline Splits

We divided our dataset chronologically:

- Train: Jan 2022 - Dec 2024
- Val: Jan 2025 - June 2025
- Test: July 2025 - Jan 2026

Why Random Cross-Validation Fails

Standard k-fold cross-validation randomly shuffles rows. This leaks future data (the target temperature at $t+k$) into the training folds, artificially inflating training scores while causing real-world models to fail. Chronological splits keep the training partition strictly in the past.

Real-World Operational Testing

Simulating Live Performance

Evaluating on a continuous, untouched out-of-sample test set (H2 2025) simulates how the models will behave in a live plant, ensuring reliability.

Validation and Early Stopping

We used the Val partition to stop tree growth when validation RMSE ceased improving, securing the model against overfitting.

EDA - Monthly & Seasonal Alarm Trends

Seasonal Ambient Influence

Yearly Alarm Frequency:

- 2022: 309 episodes | 20,162 total mins (Avg: 65.3m)
- 2023: 402 episodes | 9,742 total mins (Avg: 24.2m)
- 2024: 377 episodes | 18,749 total mins (Avg: 49.7m)
- 2025: 467 episodes | 12,638 total mins (Avg: 27.1m)

Summer vs. Winter Profiles:

Alarms peak dramatically during hot summer months (June, July, August) where the ambient air temperature degrades the heat exchange efficiency of the overhead cooling condensers.

Operational Action:

Condenser cleaning and utility cooling flow increases should be scheduled immediately prior to peak summer (May) to maximize column thermal headroom.

Yearly Alarm Accumulation Details

Year	Episodes	Total Duration	Avg Duration	Max Duration
2022	309	20,162 min	65.3 min	508 min
2023	402	9,742 min	24.2 min	353 min
2024	377	18,749 min	49.7 min	535 min
2025	467	12,638 min	27.1 min	561 min
2026 (Jan)	6	11 min	1.8 min	4 min

Model Selection & Training

Linear Regression

Description

A simple linear baseline model.

Pros: Extremely fast, easy to deploy.

Cons: Cannot capture the column's complex thermodynamic non-linear interactions.

REJECTED (Low Accuracy)

Deep Learning (LSTM)

Description

Recurrent neural network for sequences.

Pros: Excellent sequence memory.

Cons: Huge computational overhead, sensitive to missing value gaps, hard to explain.

REJECTED (High Complexity)

LightGBM Regressor

Description

Gradient Boosted Decision Tree ensemble.

Pros: Handles non-linear relationships, collinear inputs, native NaNs, ultra-fast training.

Cons: Requires careful hyperparameter tuning.

SELECTED MODEL

Hyperparameter Tuning via Optuna

Bayesian Optimization Strategy

Optuna TPE Search

Used Bayesian Tree-structured Parzen Estimator (TPE) search to locate optimal learning rates, tree sizes, and regularization parameter combinations.

1-in-8 Data Downsampling

Tuned on a statistically representative downsampled subset (187k rows) to reduce search time by 20x, then retrained on the full 2 million row history.

Regressed for RMSE

Optimized model's Root Mean Squared Error to tighten forecast curve predictions and naturally increase alarm classification precision.

Best Parameter Solutions Found

5-Minute Horizon Model Params

learning_rate: 0.141 | num_leaves: 92 | max_depth: 10 | min_child: 40

15-Minute Horizon Model Params

learning_rate: 0.148 | num_leaves: 102 | max_depth: 9 | min_child: 65

30-Minute Horizon Model Params

learning_rate: 0.034 | num_leaves: 72 | max_depth: 8 | min_child: 65

60-Minute Horizon Model Params

learning_rate: 0.047 | num_leaves: 99 | max_depth: 9 | min_child: 61

Model Performance Comparison

Optimized Precision Results

Precision Improvement across Horizons

Optuna tuning successfully increased model precision to 93.45% for 5m, 94.06% for 15m, 90.79% for 30m, and 84.89% for 60m.

Reduced False Alarm Rate (FAR)

The tuned V2 model achieves extremely low FARs: 0.208% at 5m, 0.179% at 15m, 0.278% at 30m, and 0.438% at 60m. Tighter alarms build operator trust.

Lower Regression Variance (RMSE)

Root Mean Squared Error (RMSE) is reduced at 15m and 30m, meaning predictions stay closer to actual trends under highly dynamic load shifts.

Comparative Test Set Summary Table

Horizon & Version	F1-Score	Precision	False Alarm (FAR)	RMSE
5m V1 Baseline	93.76%	93.32%	0.2129%	0.251°C
5m V2 Tuned	93.77%	93.45%	0.2079%	0.256°C
15m V1 Baseline	91.86%	93.37%	0.2026%	0.405°C
15m V2 Tuned	91.90%	94.06%	0.1790%	0.403°C
30m V1 Baseline	88.64%	90.18%	0.2996%	0.534°C
30m V2 Tuned	88.84%	90.79%	0.2783%	0.532°C
60m V1 Baseline	81.49%	84.12%	0.4707%	0.681°C
60m V2 Tuned	81.26%	84.89%	0.4376%	0.685°C

Conclusion & Next Steps

PHASE 1

Operator Simulator

Deploy the Streamlit real-time simulation tool to train control room operators, establish process workflow, and build user trust.

PHASE 2

Historian Connection

Integrate model pipeline as a microservice reading OPC UA historian live values (1-minute updates) and compute features in real-time.

PHASE 3

Operator UI Alerts

Render scrolling forecast trends on the operator console, triggering alerts 5 to 60 minutes before threshold crossing.